2.1 Introduction

Project Name	**Power Quality Improvement Using Grid-Tied Inverter and Capacitor Bank**
University	**Ahsanullah University of Science and Technology**
Department	**Department of Electrical and Electronic Engineering**
Location	**Dhaka, Bangladesh**
Duration	**3 Months**
Position	**Electrical Engineer**

2.2 Background

A. Overview

Power quality (PQ) is the stability, reliability, and efficiency of electrical power in a system. A grid-tied inverter is used to transform DC power, such as solar panels, into AC power and inject it into the main grid and grid-synchronous. Capacitor bank compensation and reactive power compensation enhance the stability of voltages and minimize the power losses. The power factor correction (PFC) makes the system efficient, and the mitigation of harmonics takes care of the distortion created by non-linear loads. Inverters and voltage source inverters (VSI) are controlled by Pulse width modulation (PWM). Active power filtering filters out the undesired harmonics. Renewable energy is supported by distributed generation (DG) integration, and the control strategies control switching losses and maximize smart grid use. All these factors combine towards efficient, stabilized, and high-quality delivery of power.

The project concentrated on the enhancement of power quality in electrical systems that were influenced by nonlinear and unbalanced loads. Initially, the quality issues of power were investigated, and appropriate compensation approaches were determined in order to minimize harmonics, reactive power provisions, and current-balancing. Next, all the components of the system were chosen thoughtfully, such as a grid-tied inverter, LCL filter, capacitor bank, and control units, which would allow the system to work effectively and reduce the harmonic distortion. Subsequently, a grid-connected inverter system was developed and simulated in MATLAB/Simulink, and control strategies were used to regulate the voltages, cancel the harmonics, and support reactive power. The system performance was evaluated by analyzing waveform improvement, power factor, and harmonic distortion before and after compensation. Lastly, the effectiveness of the proposed system was determined by the determination of a smoother operation, better power quality, and efficient use of energy.

B. Objective

The primary aims of the project were to design a grid-tied inverter system, minimize harmonics, discover power quality issues, optimize power factor, load balancing, efficiency, and model performance of the system through simulation. Its secondary objectives were: -

- To minimize the losses of power and enhance the effectiveness of the whole electrical system.
- To prevent overheating and other forms of damage to equipment due to harmonics and low-quality power.
- To facilitate the stable and reliable running of the power system during the change of the load conditions.

C. Nature of Work

I began the project by collecting data about power quality issues associated with nonlinear and unbalanced electrical loads. I researched the impact of harmonics, reactive power, and current imbalance on the electrical system. Next, I chose appropriate elements of the system, such as the inverter, LCL filter, capacitor bank, and control units, to enable successful harmonic compensations and reactive power supply. After that, I designed and simulated the grid-connected inverter system using MATLAB/Simulink, developing control strategies to improve power quality. I analyzed the system performance by checking current and power waveforms, power factor, and total harmonic distortion before and after compensation. Finally, I concluded the system’s effectiveness, confirming that it improved power quality, reduced harmonics, increased efficiency, and ensured efficient and stable operation.

D. Hierarchical Structure

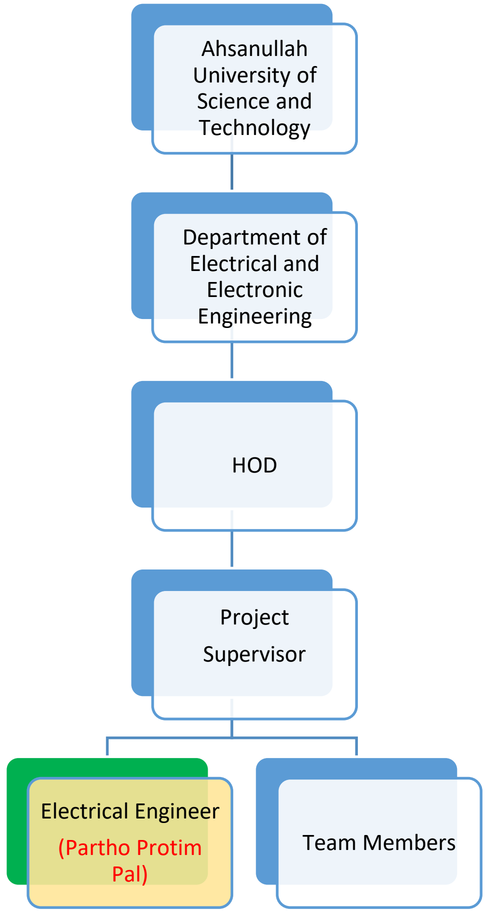

Figure 1: Organizational Hierarchy

E. Duties and Responsibilities

- To study the issues of power quality associated with nonlinear and unbalanced electrical loads and find appropriate ways of compensation.
- To choose the appropriate system elements, such as inverter, filter, capacitor bank, and control units to be able to provide the harmonic compensation and the reactive power support most efficiently.
- To design and simulate a grid-connected inverter system in MATLAB/ Simulink with effective control strategies to improve the power quality.
- To assess the system performance in terms of the improvement of the waveforms, weight factor, and harmonic distortion prior and after the compensation.
- To summarize the effectiveness of the system in enhancing the power quality and the stable and efficient operation.

2.3 Personal Engineering Activities (PEA)

A.

I began this project by researching the issue of poor power quality in electric systems. First, I determined that harmonics, reactive power demand, and current imbalance were due to nonlinear and unbalanced loads like rectifiers, motor drives, and electronic equipment. I realized that these issues led to the compromise of power, overheated equipment, low efficiency, and low power factor. I studied the fundamentals of power quality and harmonics, the working of an inverter, and the compensation of reactive power. I applied my engineering background in electrical systems, mathematics, and power electronics to learn how the problems impacted the grid. Subsequently, I learned about the concept of applying a Multi-Functional Grid-Tied Inverter and a capacitor bank to address these issues. I discovered that the inverter was capable of minimizing harmonics, reactive support, and load balancing, and the capacitor bank could minimize the reactive load on the inverter. I also used safety, standards, and sustainable engineering practice when choosing this solution. By this phase, I managed to form a clear vision of the problem and find the appropriate technical approach and prepared the foundation for the design and simulation work.

B.

I chose system requirements and selected the system elements in accordance with the project goals and requirements. To begin with, I chose a 400 V DC voltage source since the inverter required a DC link that would be constant in order to work properly. Next, I decided to use a three-phase supply at 400 V, 50 Hz grid supply since it corresponded to a typical practical power system. Then, I decided to employ an IGBT-based voltage source inverter as it was the fastest switching and best-controlled tool to achieve harmonic compensation. To reduce switching ripple, I chose to add an LCL filter, L_1 = 3 mH, L_2 = 2 mH, and C = 20 µF, since it enhanced the quality of the current prior to the inverter current being on the grid. I also chose a capacitor bank since it provided some of the reactive power and lessened the reactive power load on the inverter. To the load side, I selected an R-L load as a linear load and a diode bridge rectifier with a capacitive filter as a nonlinear load due to the representations of realistic industrial scenarios. I chose a PWM generator, a PI controller, and a current controller as well to switch and control. Lastly, I employed measurement blocks and Current Measurement blocks, power measurement instruments, FFT analysis instruments, and MATLAB scripts since I required a precise performance of monitoring, waveform examination, and THD analysis during the project.

C.

I created the system in MATLAB/Simulink to be a grid-connected compensation system. I linked the inverter to the load and grid with an LCL filter in such a way that the inverter would inject compensating current at the Point of Common Coupling. I then organized the system into distinct parts: grid supply, inverter unit, LCL filter unit, load section, capacitor bank, and control section. I then examined the working principle carefully. I noticed that the nonlinear load possessed distorted current and reactive power on the source. On this basis, I adjusted the inverter to produce offsetting current to eliminate harmonic content and to supply reactive power demand. I also gave the capacitor bank the option of supplying a portion of reactive current in order to make the inverter more effective. I then proceeded to develop the control process step by step. I measured voltages and currents at the PCC, removing unwanted harmonic and reactive components, creating a reference current, and applying PWM switching to regulate the output of the inverter. I also applied a PI controller to control the DC-link voltage. Lastly, I checked the design both in MATLAB scripts and by Simulink simulations to confirm that the system propagated under real operating conditions. I adopted MATLAB programs to produce harmonic signals, device power factor, and theoretical harmonic phase. I also simulated the inverter and grid interaction using Simulink. Through this work, I utilized engineering problem-solving, designing, the use of tools, and systematic project management skills.

D.

I assessed the performance of the system by comparing the outcomes prior to and post-compensation. I investigated the existing waveform and discovered that the waveform was distorted immensely prior to compensation due to nonlinear loads. After using the inverter and capacitor bank compensation, I measured that the current waveform was smooth and almost sinusoidal. This demonstrated that the harmonic mitigation technique was effective. I then computed the instantaneous power and discovered huge oscillations prior to being compensated. On applying the compensation, the power waveform was smooth, and the fluctuations were minimized. I also examined the power factor and observed that the power factor before compensation was approximately 0.75-0.85, and after compensation was almost 0.98, which offered an indication of almost unity power factor operation. Then I performed FFT analysis and discovered that the total harmonic distortion decreased by more than 25–35% to 3-5 %. I also noticed that the high 3rd and 5th harmonics were lowered considerably after compensation. Based on these, I concluded that I used engineering analysis, interpretation of data, and performance evaluation well, and I presented the findings clearly using a waveform plot, FFT findings, and summary comparisons.

E.

I concluded the project by ensuring that the proposed system achieved its objectives of improving power quality through the use of a grid-tied inverter and a capacitor bank. First, I checked that the harmonic distortion was minimized, which was related to the improved quality of the current waveform and decreased the adverse impact of nonlinear loads on the grid. Next, I ensured that the power factor increased to almost unity, which implied that the system consumed electrical power more effectively. After that, I observed that smoother values of the current and power waveforms displayed improved system performance. I also realized that the capacitor bank was also significant as it provided a section of reactive power load, which decreased the load on the inverter and helped to maintain the stable operation. Through this project, I showed a skill to implement the basic engineering principles, choose appropriate components, employ simulation tools, interpret findings, and to address a broadly-defined engineering problem. I also planned the safety, efficiency, and sustainable engineering practice when working. Also, I enhanced my professional competence by structuring the project systematically and clearly presenting the findings. Generally, I concluded that this system was useful and would be applicable in industrial and commercial settings where power quality enhancement was needed. It also increased my confidence in the modern engineering approach to address real-world electrical technology problems simply and efficiently.

2.4 Technical Problems and Solutions

A.

During the planning phase of the project, I encountered a technical issue of bad power quality in the system. I noticed that the present waveform was so distorted, and the power factor was poor, which adversely impacted the system efficiency and losses. I found that nonlinear loads, such as rectifiers and electronic equipment, caused harmonics and reactive power demand. The problems also caused overheating of equipment and unstable operation. I read and studied various types of compensation systems and chose a grid-tied inverter with a capacitor bank. To overcome this, I installed the inverter in order to inject matched current and a capacitor bank to provide reactive power. Upon the problem solution implementation, I noticed that the present waveform was smooth and minimized harmonics, and the power factor greatly increased.

B.

Another technical issue I faced was the high Total Harmonic Distortion (THD) and current ripple between the parts of the grid connection. I observed that ripple was produced by the inverter switching that influenced the quality of the current and lowered system performance. This issue adversely affected the precision of compensation and caused additional workloads on the electric elements. I researched the cause of this and discovered that the harmonics could be switched into the grid by improper filtering. I learned the various filter designs and used an LCL filter to improve the performance. To overcome this, I developed and applied an LCL filter having appropriate parameters. Following the implementation, I noticed that the ripple was less, THD was low, and the overall system performance was enhanced.

2.5 Creative Work

I identified problems of power quality and studied their causes. I chose and used a grid-tied inverter, capacitor bank, and LCL filter. I also designed solutions, testing them and enhancing the system performance in terms of minimizing harmonics and ripple, and increasing power factor.

2.6 Team Coordination

I led the project team by initially giving roles and responsibilities to the various team members in accordance with their skills. I coordinated regular meetings to discuss progress, challenges, and next steps, ensuring everyone understood the project goals and deadlines. I directed the team members throughout system design, selecting components, setting up the simulation, and assisting the team with technical aspects and clarifying questions. I supervised the work, checked the MATLAB/Simulink models, and made sure that each task was carried out correctly and punctually. I also promoted teamwork and open communication to ensure that issues were resolved promptly. I also recorded team activities, updates, and organized testing and performance evaluation. In this process, I ensured a well-organized workflow, accountability, and the team managed to deliver successful outcomes in enhancing power quality through the grid-tied inverter and capacitor bank.

2.7 Codes and Standards

I created MATLAB/Simulink code for the inverter, the PWM, the PI controller, and the measurement tools, and I adhered to the IEEE 519 and IEC 61000 standards of power quality limit and harmonics, which was a safe, accurate, and standard-compliant power quality enhancement.

2.8 Summary

A.

This project involved the enhancement of the quality of power in electrical systems impacted by nonlinear and unbalanced loads. The work was initiated by the analysis of such issues as harmonics, the demands of reactive power, and an imbalance of currents, and finding appropriate ways of compensation.

Appropriate system components were selected, including a grid-tied inverter, LCL filter, capacitor bank, and control units, to provide effective harmonic mitigation and reactive power support. A grid-connected inverter was developed and modeled in MATLAB/Simulink with control schemes to enhance the current waveform and power factor. The performance of the system was tested using the waveforms, harmonic distortion, and power factor before and after compensation. The findings indicated a large decrease in the harmonics, smoother waveforms, as well as a close to unity power factor. Lastly, the project determined that the suggested system was effective in enhancing the power quality and providing efficient and stable functioning in practical use.

B.

I enhanced my interpersonal skills by working with others and demonstrating technical outcomes. I developed my project management abilities through the organization of tasks, simulation planning, and efficient work organization. I learned and understood more about power quality, inverters, reactive power, and harmonic compensation.

Appendix

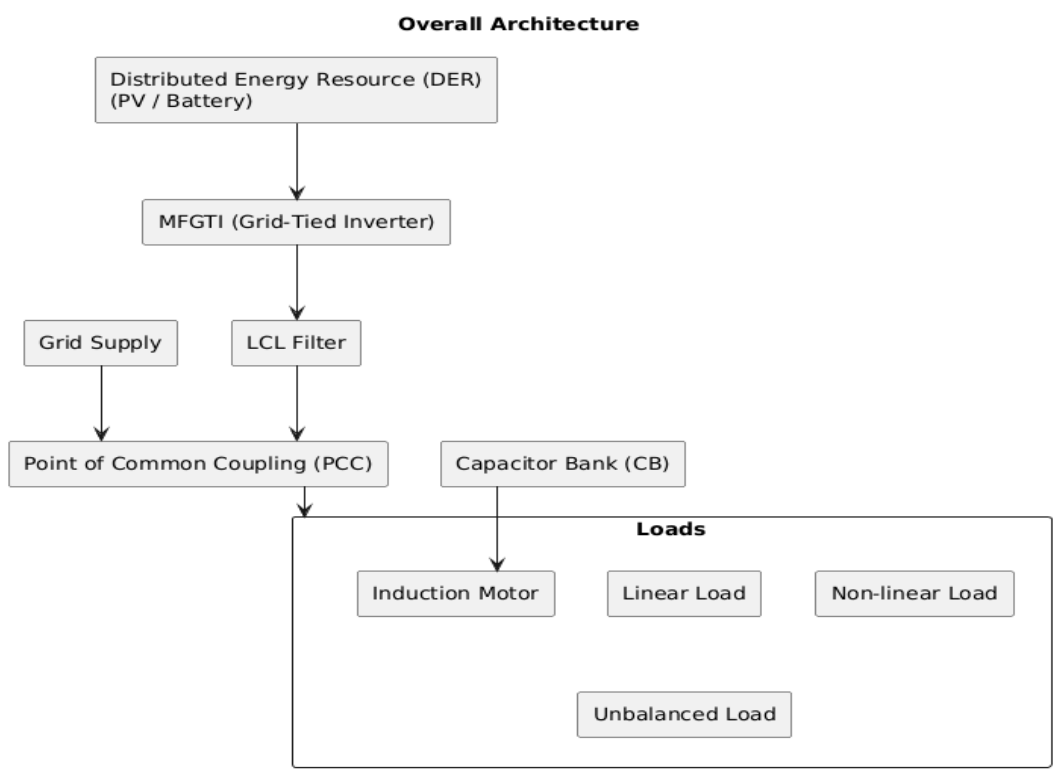

Figure 2: Overall Architecture

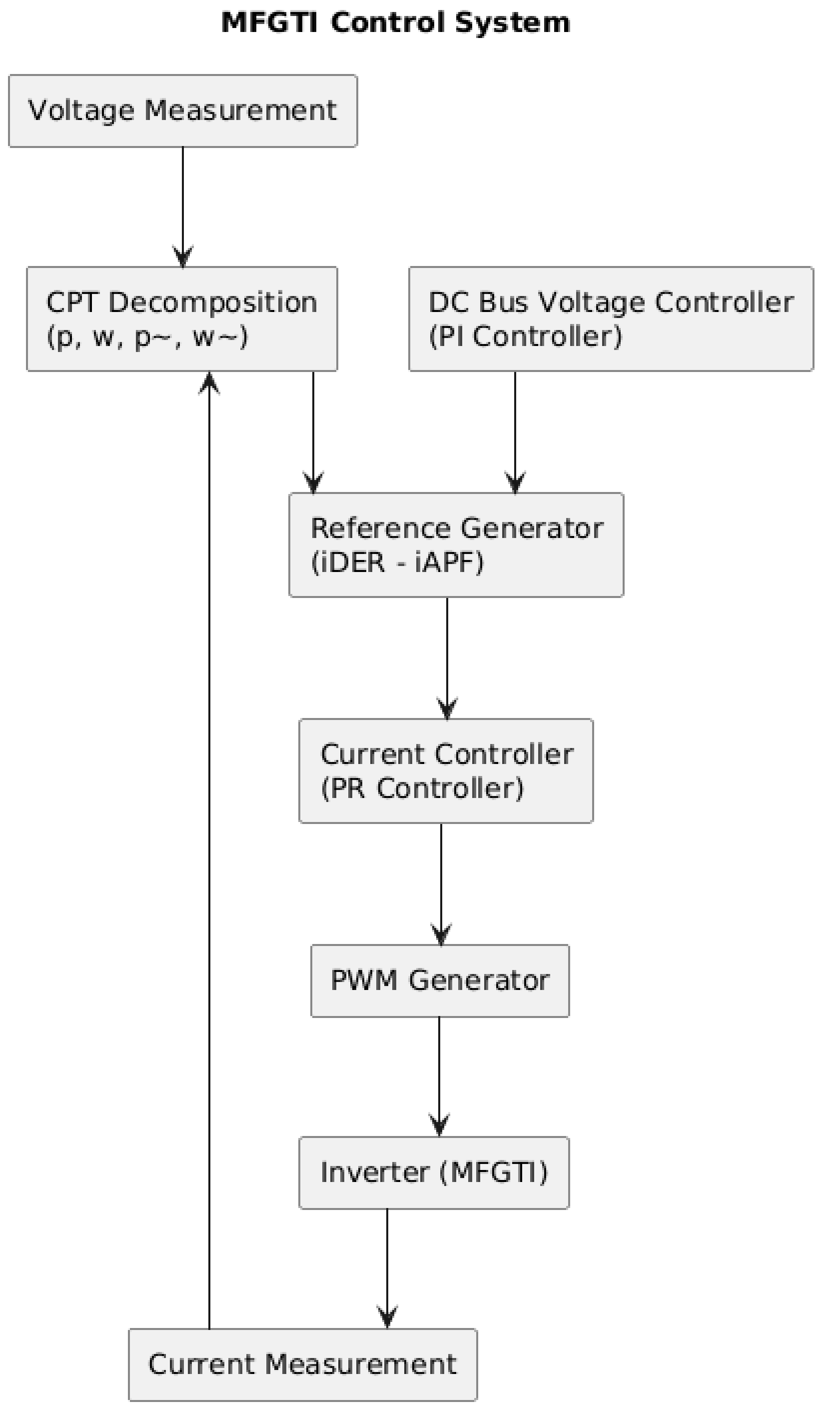

Figure 3: MFGTI Control System

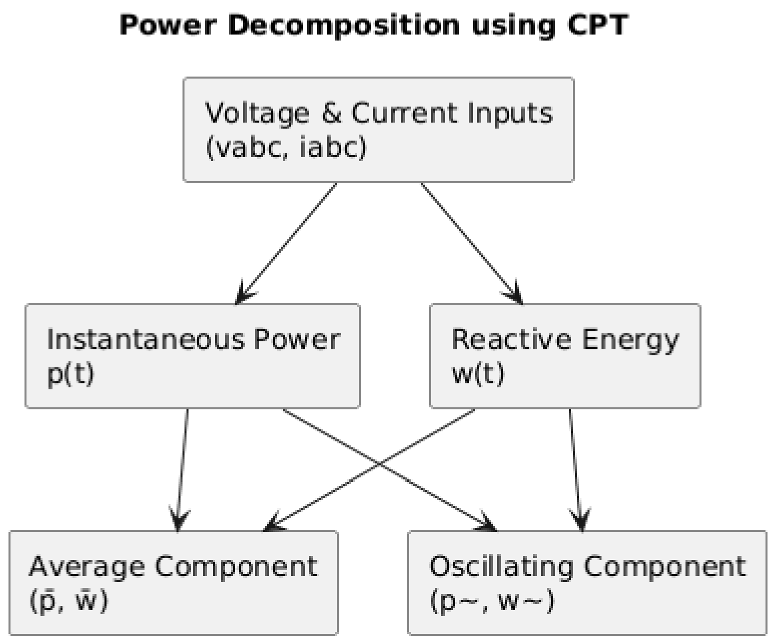

Figure 4: Power Decomposition using CPT

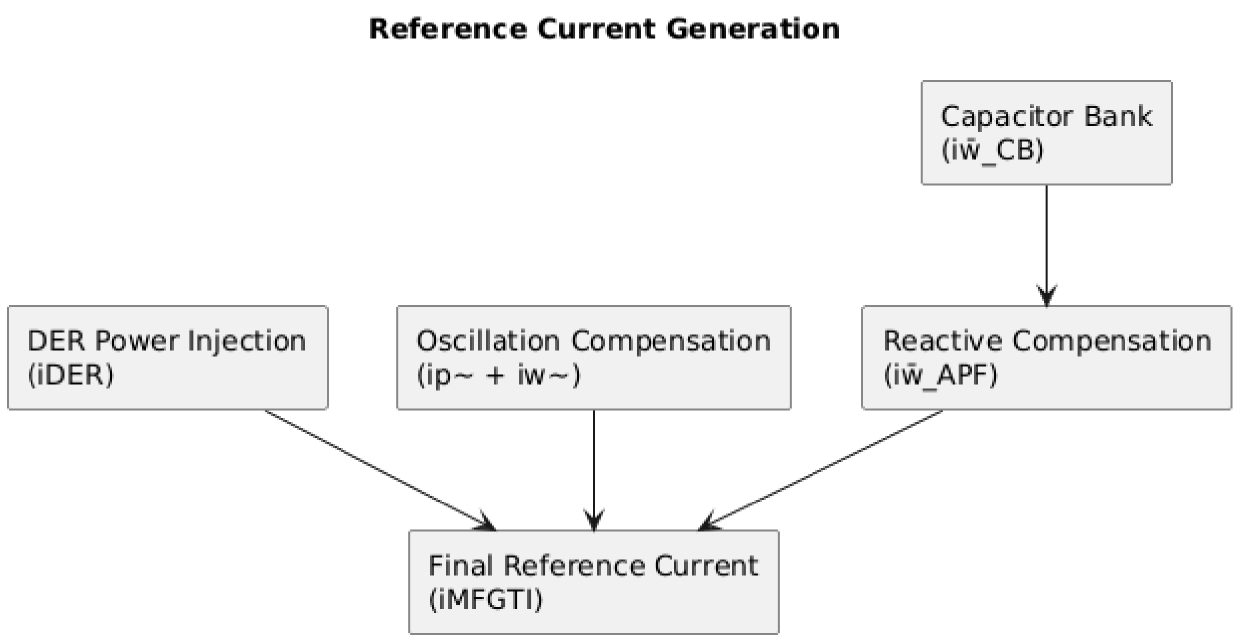

Figure 5: Reference Current Generation

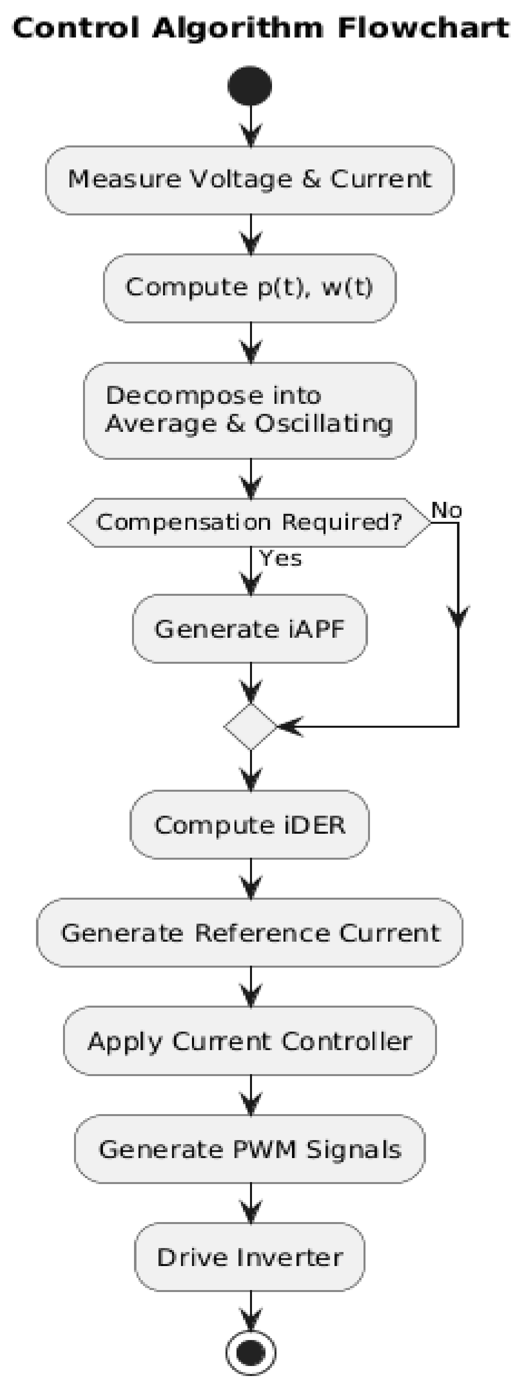

Figure 6: Control Algorithm Flowchart

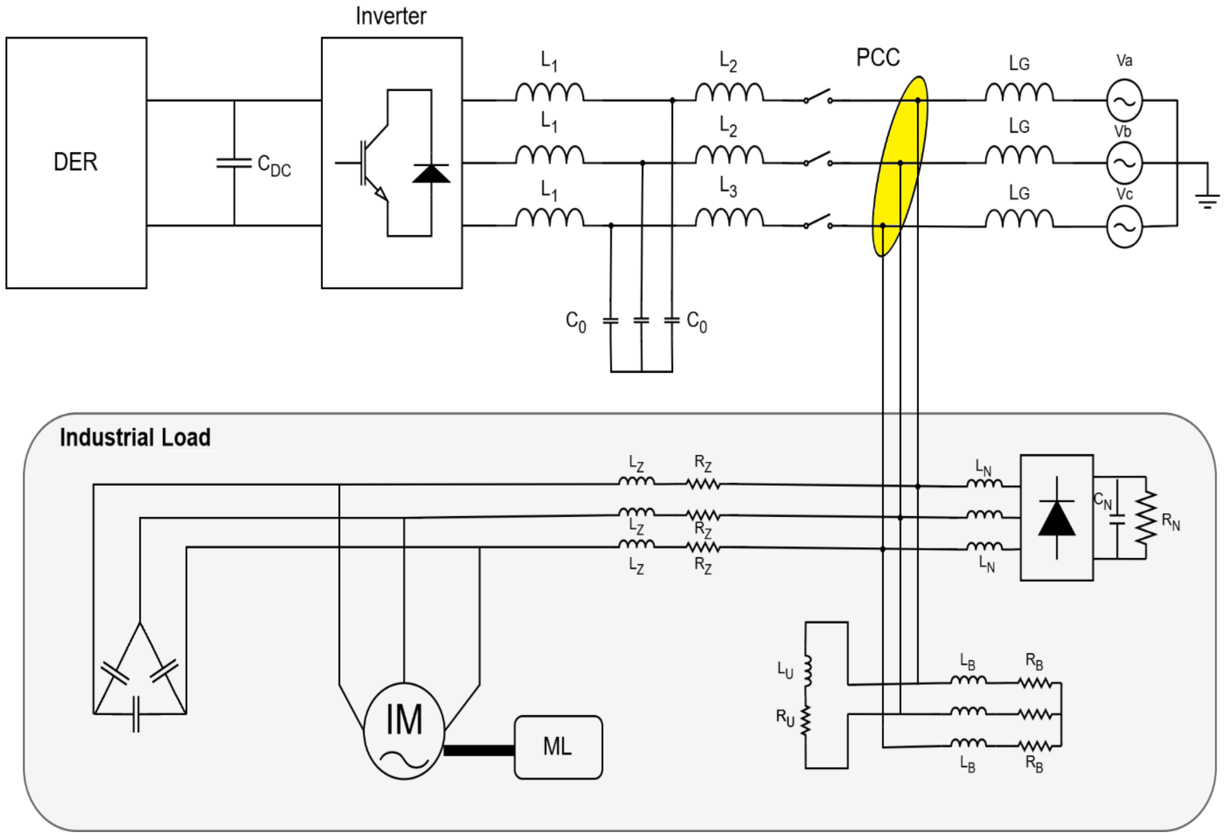

Figure 7: Circuit Diagram

Table 1: Results Summary Table

Parameter	Before Compensation	After Compensation
THD (%)	25–35%	3–5%
Power Factor	0.75–0.85	~0.98
Current Shape	Distorted	Sinusoidal
Power Quality	Poor	Improved

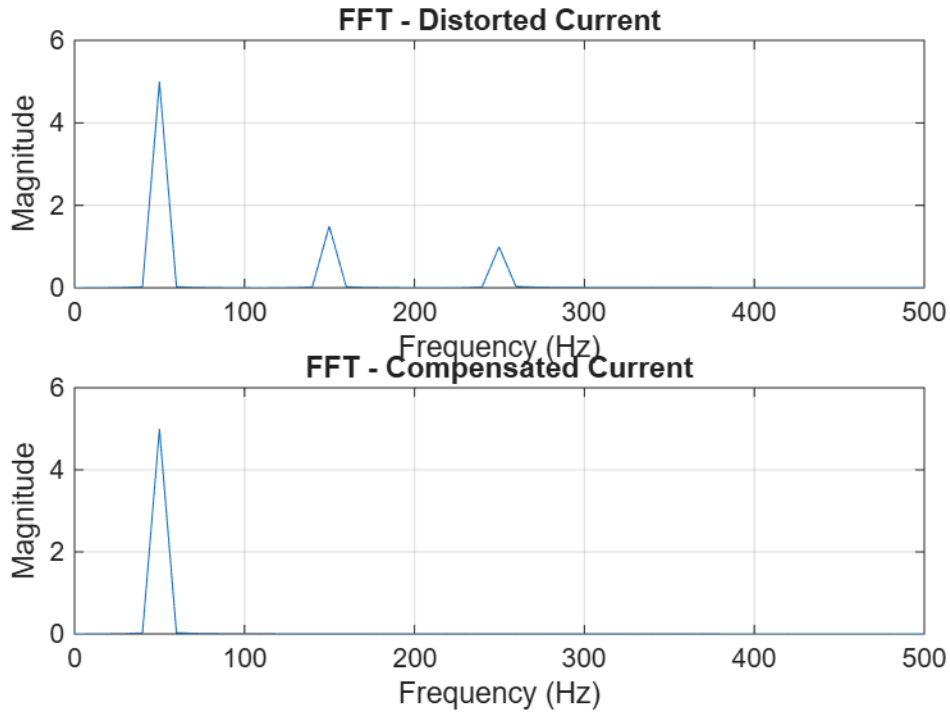

Figure 8: FFT_Analysis

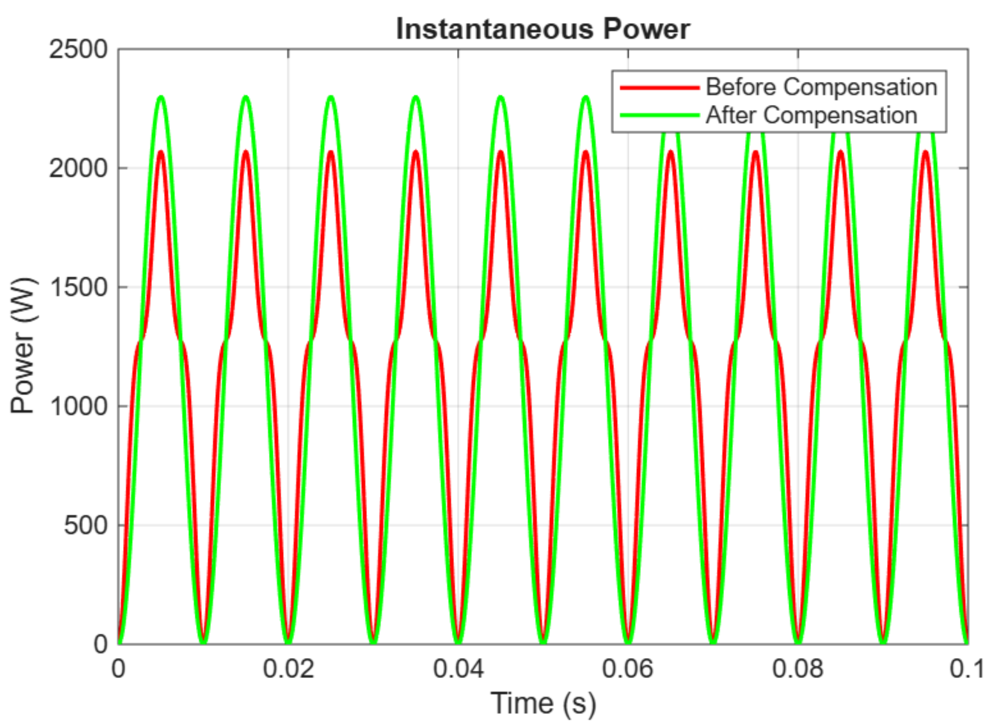

Figure 9: Power Comparison

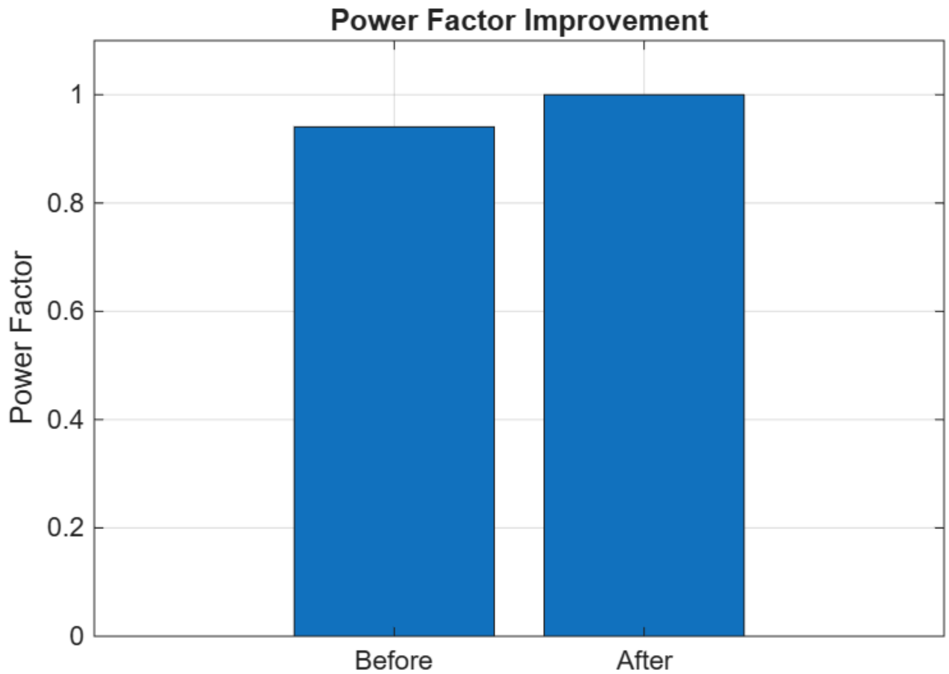

Figure 10: PowerFactor_Comparison

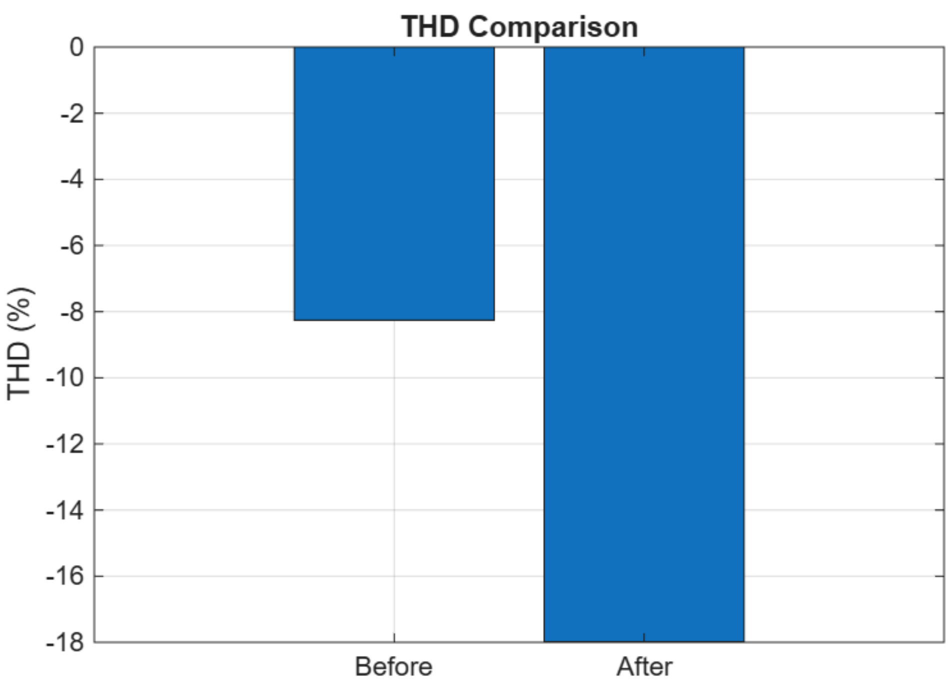

Figure 11: THD_Comparison

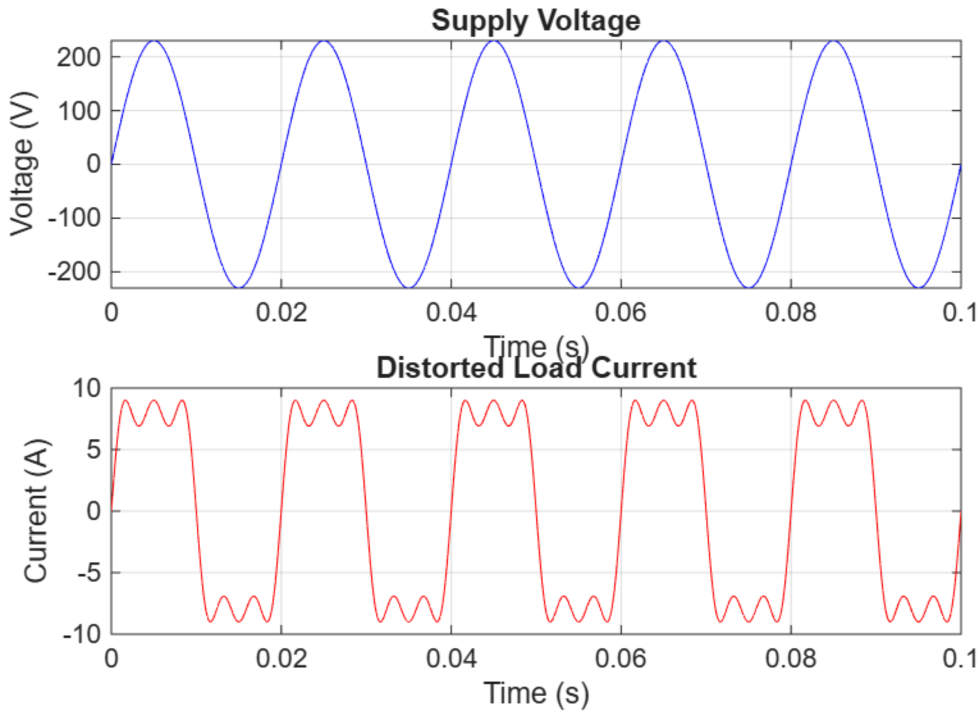

Figure 12: Voltage Current